

Unit 1 Lesson 1

makeblock

The Secret of Codey Rocky



Inquiries of this lesson

- What is a program and what can the program do
- What is Codey Rocky
- What is mBlock 5
- How to upload your first program to Codey Rocky through mBlock 5

CSTA: 1B-CS-01; 1B-CS-02



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Warm Up

What is Codey Rocky?



Codey Rocky

- **A tiny but versatile robot;**
- **Help learn programming;**
- **Display various animations and facial expressions;**
- **Avoid obstacles;**
- **Follow lines; etc.**

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Do you know these robots?



Robots in Logistics

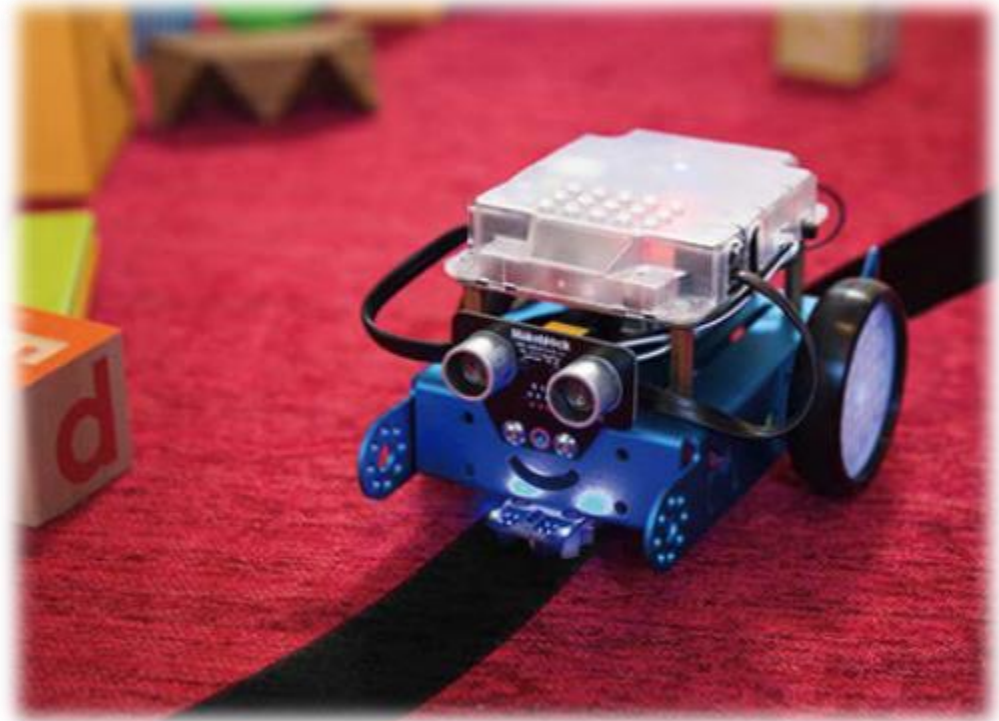


Security Robot

Do you know these robots?



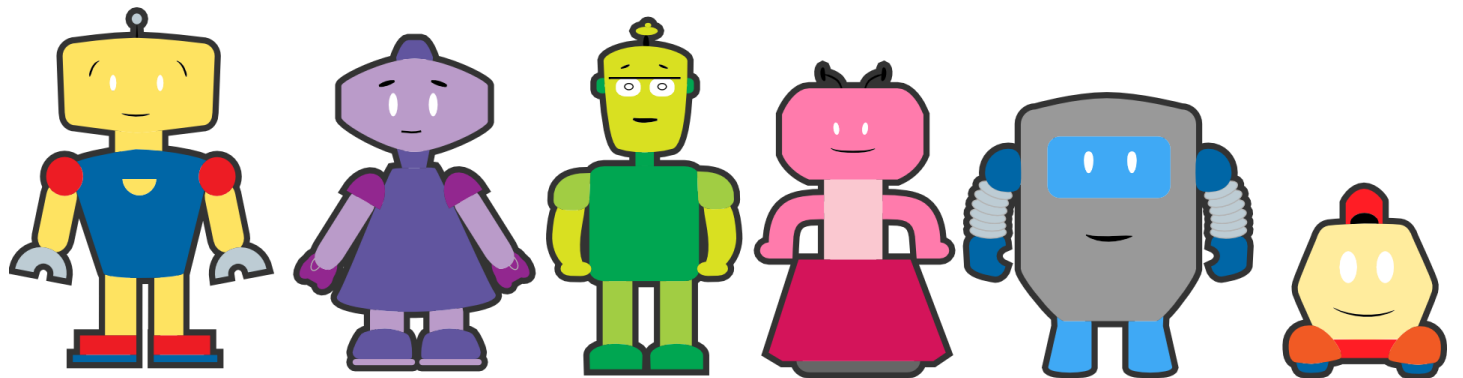
Delivery Robot



Educational Robot

Discuss:

- *Are robots alive?*
- *How can these robots understand our instructions?*
- *How can these robots carry out tasks such as food and good delivery?*



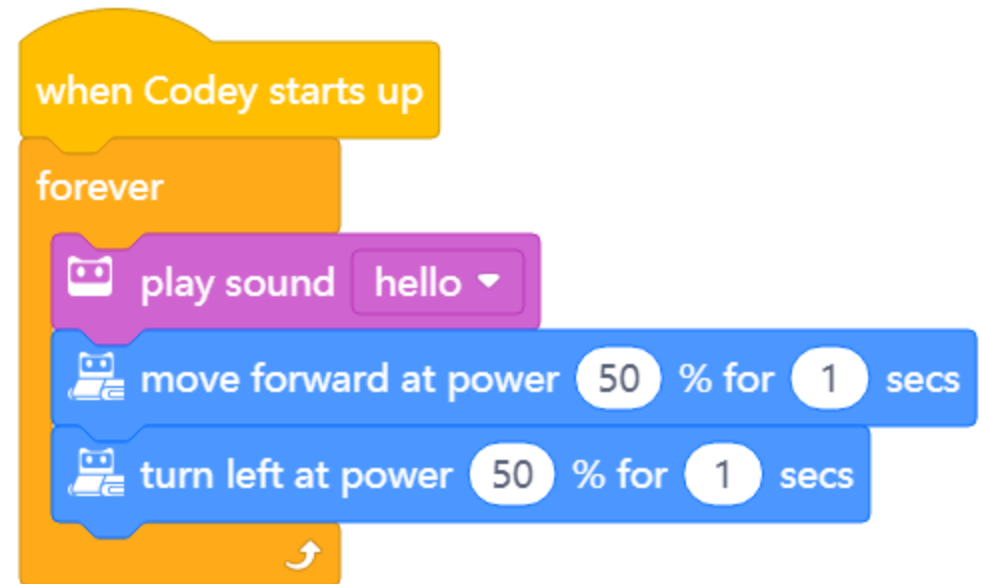
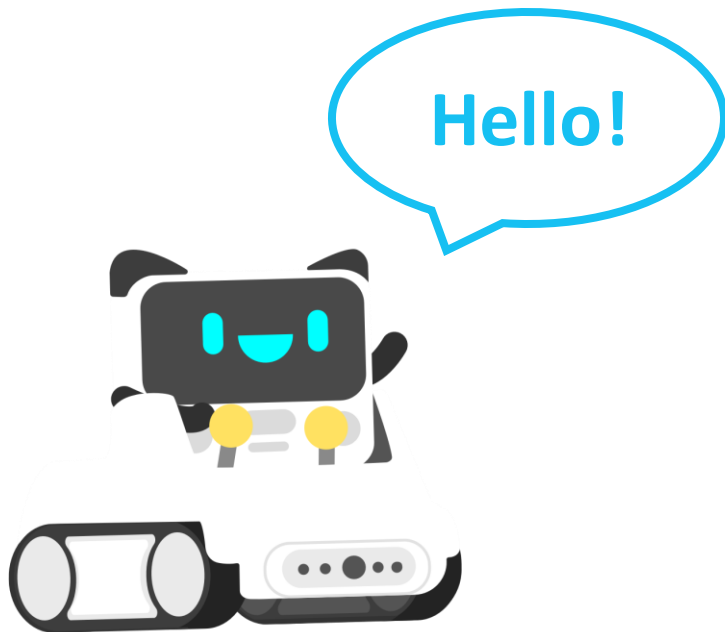
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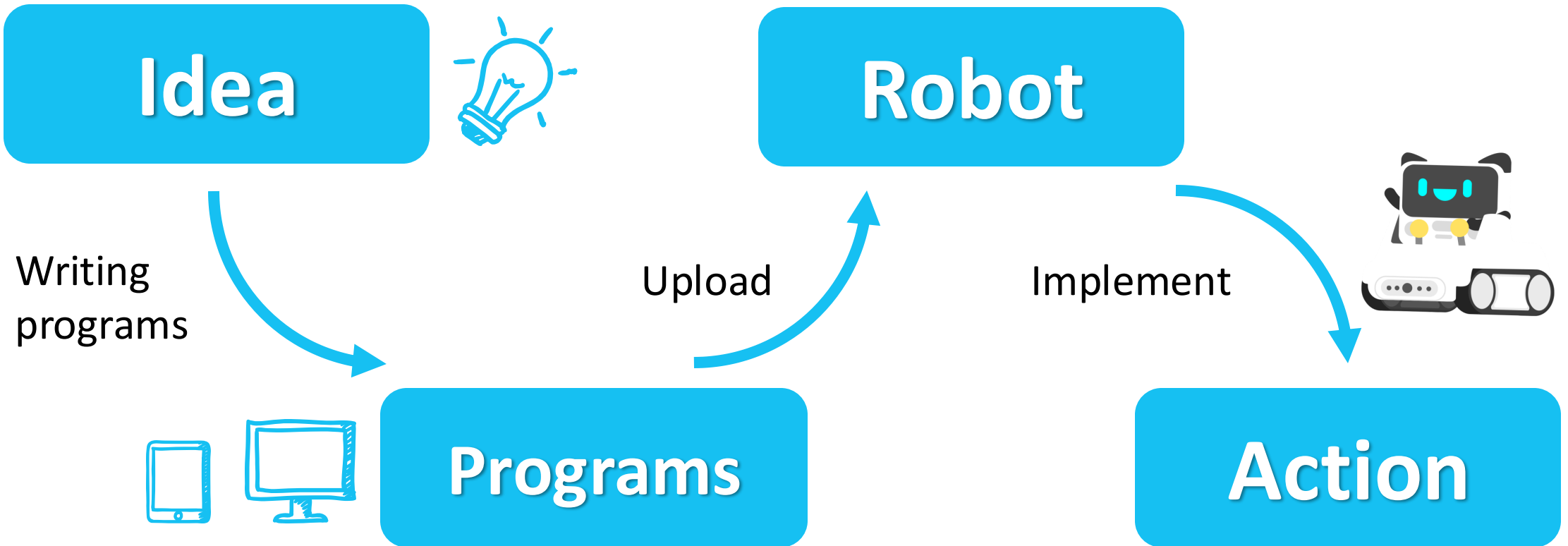
New Concept

Program

- Program is an artificial language that we use to tell robots what to do.



Program



Structure of Codey Rocky



Cody



Rocky



Codey Rocky

Features of Codey



Features of Rocky



IR Color Sensor

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Coding Practice

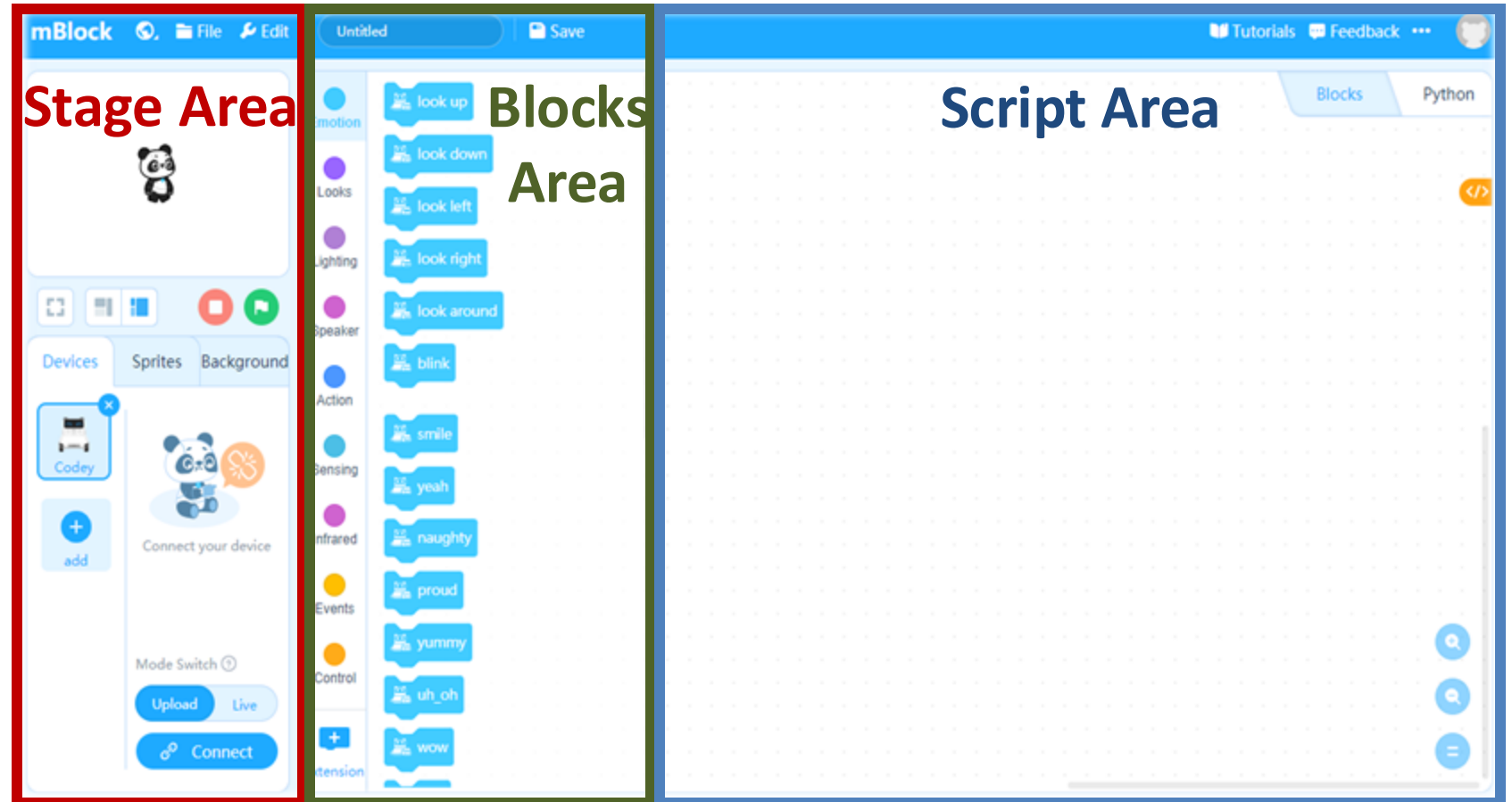


The Mblock 5 Editor

It is a visual programming platform used to code and upload projects to robots like Codey Rocky.

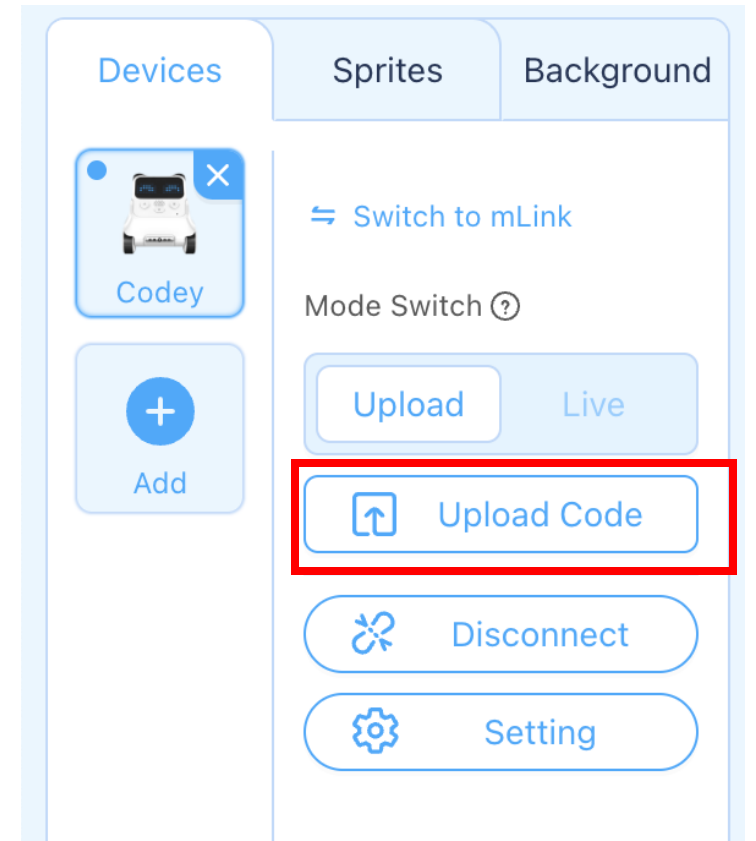
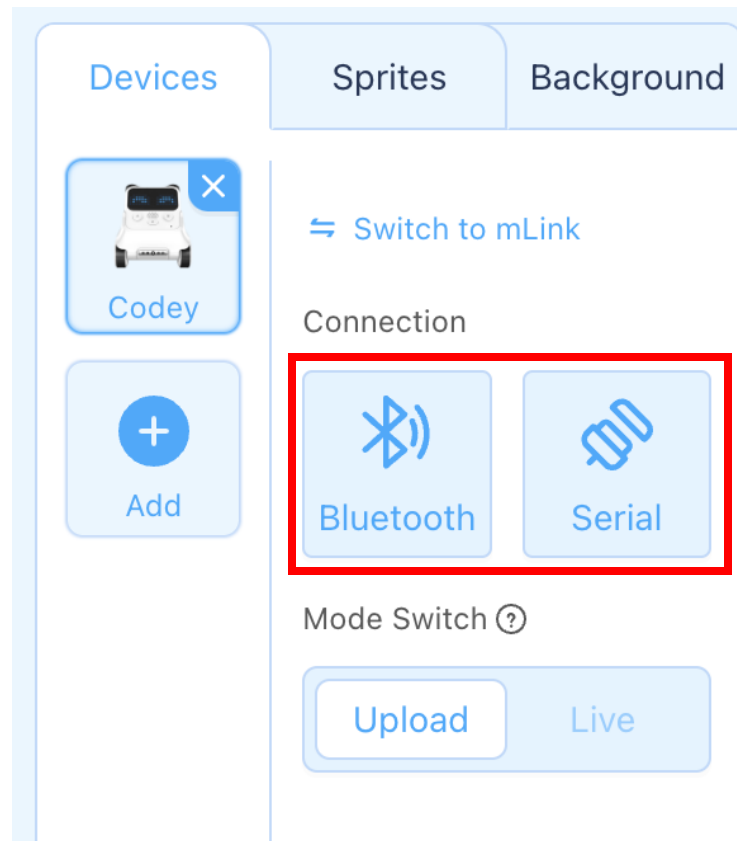
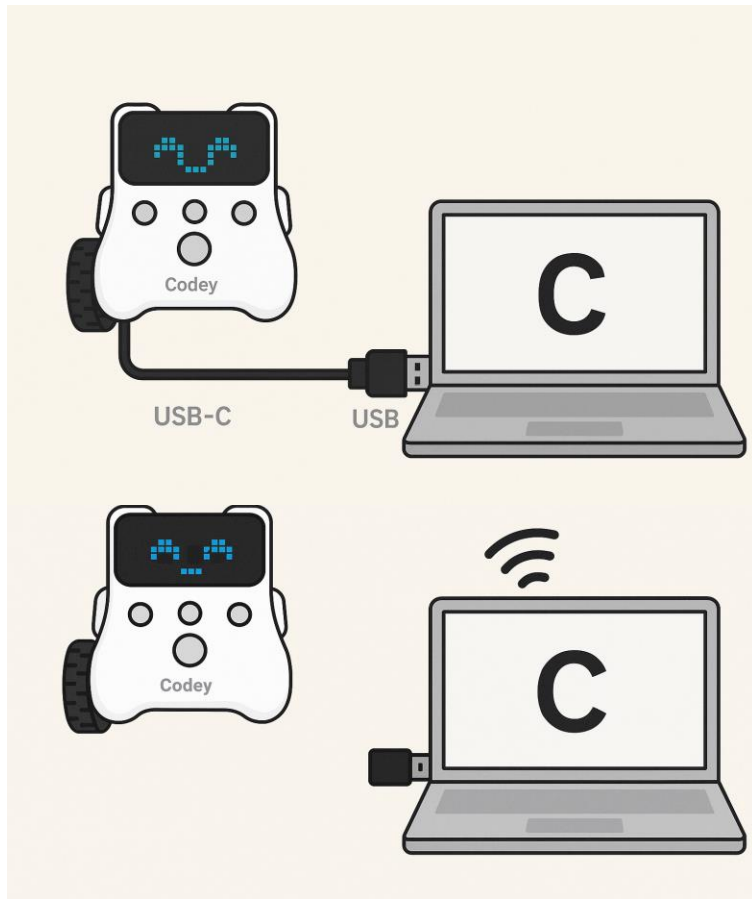
Its interface is divided into three main areas:

- The Stage Area: Shows the actions of your sprites and devices. You can connect your robot, switch modes, and preview results.
- The Blocks area: Contains all programming blocks, organized by categories like Motion, Events, and Control.
- The Script area: The workspace where you drag and snap blocks together to build your program.



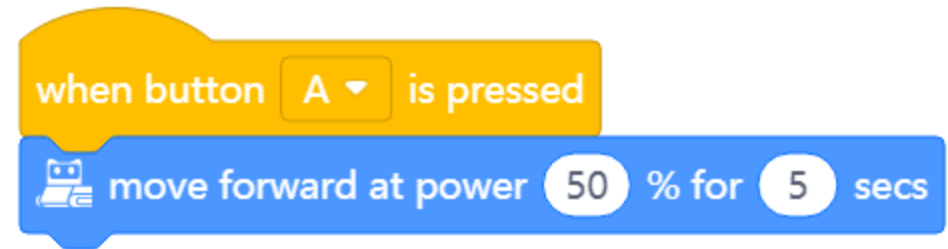
Uploading Your Program to Codey Rocky

1. Connect your Codey Rocky to the computer via the USB Cable or the Bluetooth dongle.
2. Choose the Bluetooth or Serial (USB) connection and connect your device using the device name or the corresponding USB.
3. Click on the “Upload Code” button on the Mblock editor.



Your First Executable Program

- Make and upload the code shown below to your Codey Rocky



- Modify the program, e.g., change the **power** or the **second**; etc.

Your Second Executable Program

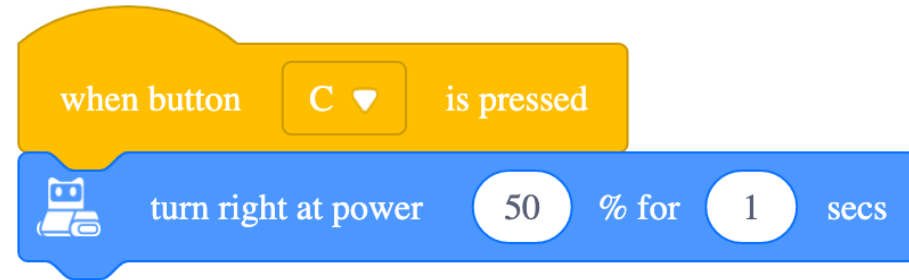
- Make and upload the code shown below to your Codey Rocky



- Modify the program, e.g., change the **power** or the **second**; etc.

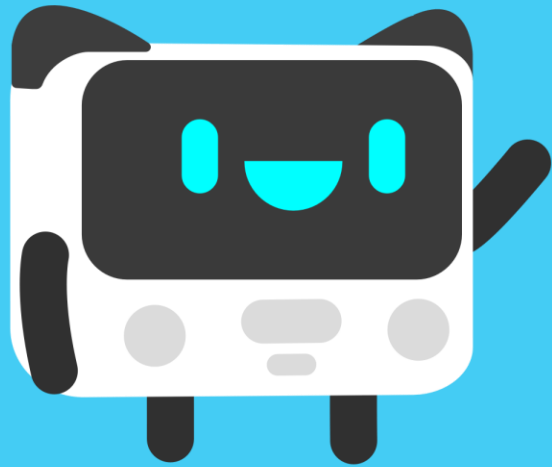
Your Third Executable Program

- Make and upload the code shown below to your Codey Rocky



- Modify the program, e.g., change the **power** or the **second**; etc.

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Wrap Up

What we've learned today

- **Program:** Program is an artificial language that we use to tell robots what to do. We translate instructions into a piece of program and upload the program to the robot, making it do a variety of things as programmed.
- **Codey Rocky** and **mBlock**

